Laboratory work 10

CASE STUDY: ANGULAR VELOCITY CONTROL OF A DC MOTOR

The experimental equipment can be found in Fig. 1. The entire unit is equipped with sensors for measuring the rotational speed of the DC motor (leftmost component). To control the DC motor a voltage signal is applied ranging from [-12, +12]V and scaled in a [-1, +1] range. The experimental unit has several components attached to create friction, inertia, etc.

You can find additional information in the dedicated folder of the laboratory (Desktop-> IRA1_LAB_SERVO).

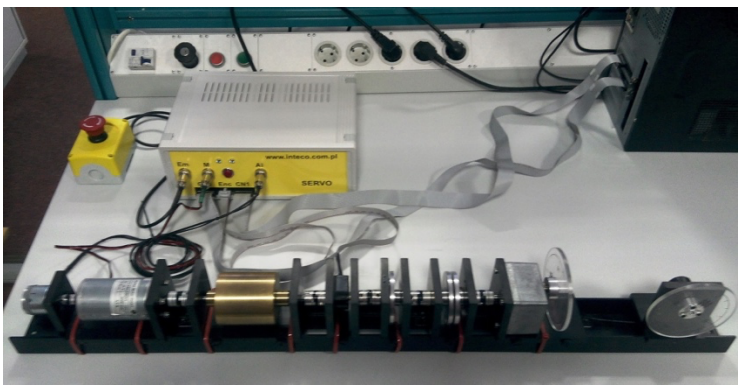


Fig. 1. Modular servo system

The mathematical model is given by the equation:

$$H_f(s) = \frac{y(s)}{u(s)} = \frac{k}{Ts+1} \tag{1}$$

where y is the rotational speed of the DC motor, u is the control signal (voltage applied to the DC motor, scaled in the [-1,+1] range, k and T are the process gain and time constant, respectively.

Different PID controllers will be tested for the control of the velocity of the DC motor. The closed loop step response will be analised in each case with respect to the following performance specifications: overshoot, settling time, steady state error to a step reference signal.

IRA1_LAB_SERVO and Velocity_ctrl_IRA1.mdl.

The following form of the PID controller is implemented in the Simulink PID block:

$$H_c(s) = k_p + \frac{k_i}{s} + \frac{k_d N s}{s+N} \tag{2}$$

where k_p, k_i and k_d are the proportional, integral and derivative gains. N is the filter coefficient and it has been preset to 10.

To determine the overshoot, settling time and steady state error the following figure and equations are used:

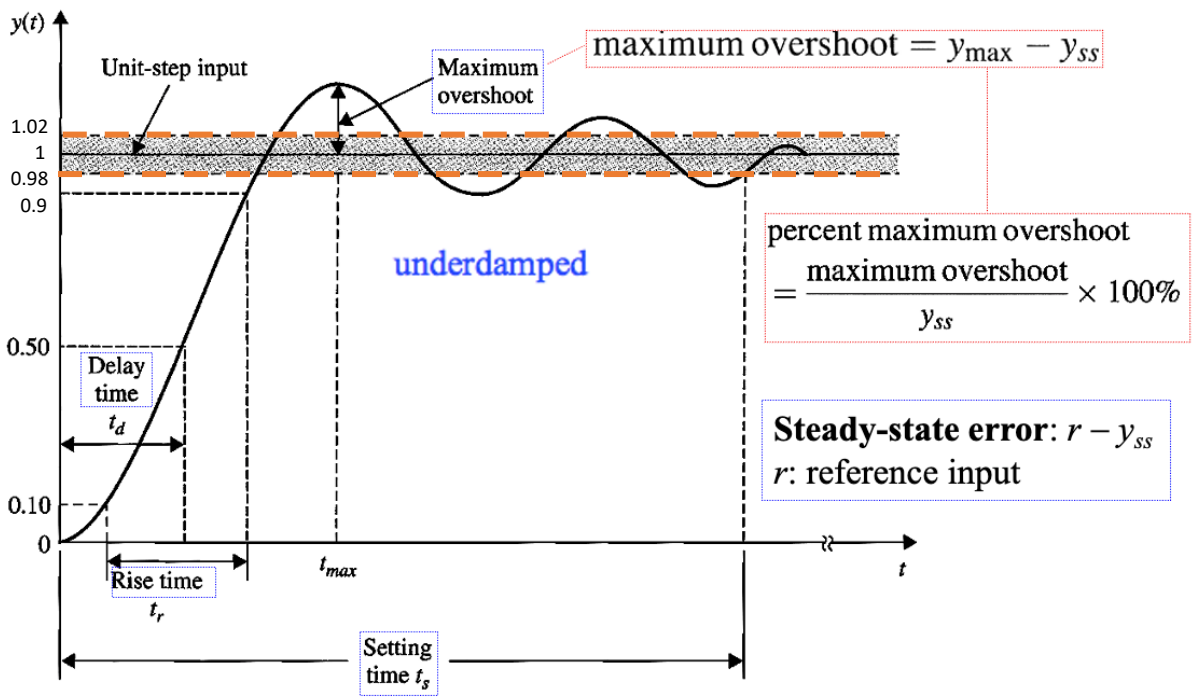


Fig. 2. Performance specifications in the time domain

$$\sigma = \frac{y_{max} - y_{ss}}{y_{ss}} 100\% \tag{3}$$

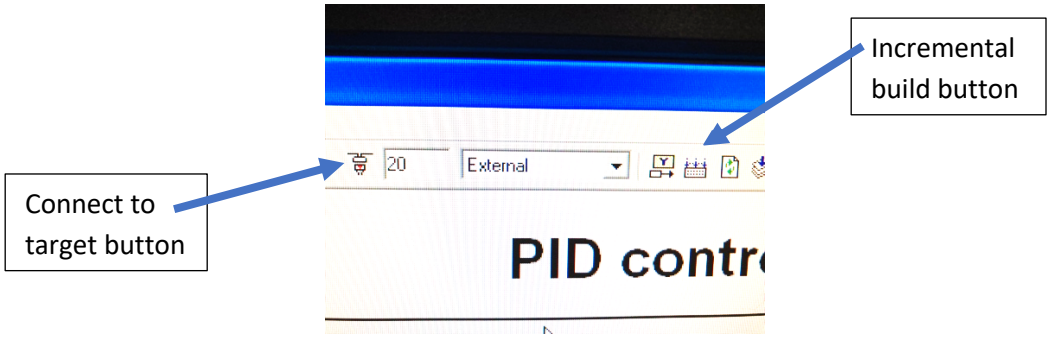
$$\varepsilon_{ss} = r - y_{ss} \tag{4}$$

where r is the reference signal, y_{max} is the maximum amplitude of the output signal and y_{ss} is the steady state value of the output signal.

To determine the settling time, the 2% criterion is used. First compute $1.02 \cdot y_{ss}$ and $0.98 \cdot y_{ss}$. Place this lines on the output plot as indicated in Fig. 2 (see red dashed lines). Determine the moment in time when the output signal enters the $[0.98 \cdot y_{ss},\ 1.02 \cdot y_{ss}]$ and remains inside this range. This will result in the settling time.

Test the following controllers and write down the results regarding the performance specifications.

Controller	Steady state error	Overshoot	Settling time
P (kp=0.01, ki=0, kd=0)			
P (kp=0.05, ki=0, kd=0)			
PI (kp=0.05, ki=0.1, kd=0)			
PI (kp=0.05, ki=0.5, kd=0)			
PID (kp=0.05, ki=0.5, kd=0.005)			

PID (kp=0.05, ki=0.5, kd=0.01)			

The Simulink file uses the real time target. To run the simulink file, first click on incremental build. Once the simulink diagram has been successfully loaded on the device, then click on connect to target and run. A successful loading of the controller can be verified in the Matlab workspace: "Successful completion of the Real Time Workshop…".